

# Servo & Tracking – Worked Problems

Practice: integral servo design and steady-state tracking limits.

Ogata, *Modern Control Engineering*, Ch. 10 (servo systems).

Step through with **Ctrl+Enter**, or render a report with **publish**.

## Contents

- Problem 1: Integral servo to a pole-placement spec
- Problem 2: what a type-1 servo can and cannot track
- Try it yourself

## Problem 1: Integral servo to a pole-placement spec

For  $G(s) = \frac{1}{(s+1)(s+3)}$ , design integral-plus-state feedback that places the augmented poles at  $-4, -5, -6$ , and confirm zero steady-state error to a step **and** rejection of a constant input disturbance.

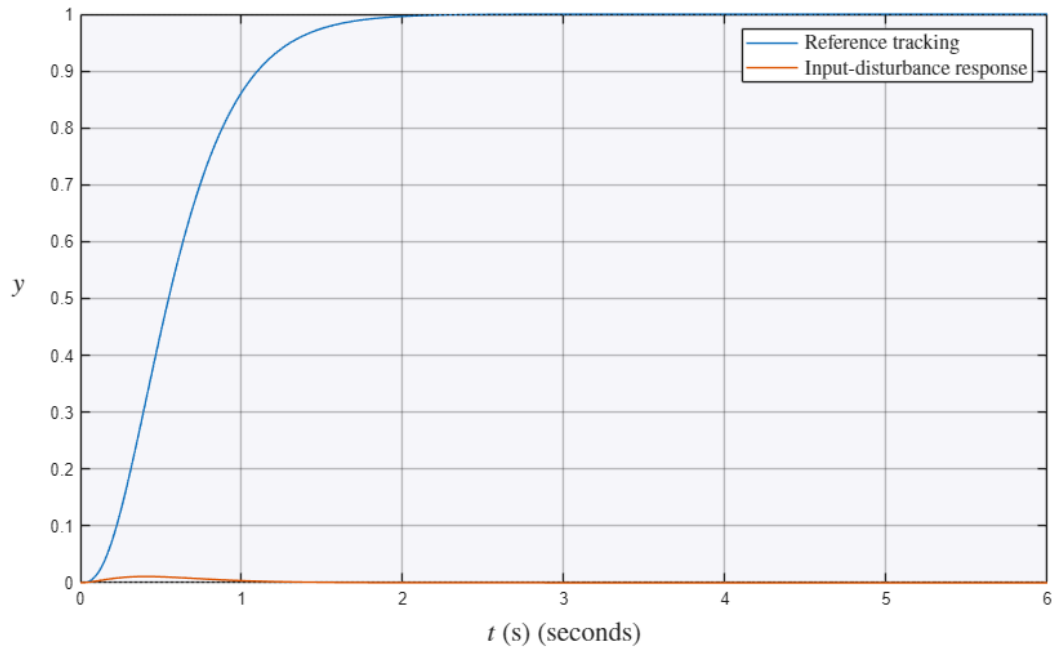
```
A = [0 1; -3 -4]; B = [0;1]; C = [1 0];
Aa = [A, zeros(2,1); -C, 0]; Ba = [B;0];
Ka = place(Aa, Ba, [-4 -5 -6]);
K = Ka(1:2); Ki = Ka(3);
Acl = [A - B*K, -B*Ki; -C, 0]; Ccl = [C, 0];

sys_r = ss(Acl, [0;0;1], Ccl, 0); % reference -> output
sys_d = ss(Acl, [B;0], Ccl, 0); % input disturbance -> output
fprintf('Problem 1: step tracking error = %.2e, disturbance DC gain = %.2e\n', ...
    1 - dcgain(sys_r), dcgain(sys_d))

t = 0:0.01:6;
figure
step(sys_r, t)
hold on
step(sys_d, t)
ylabel(1,'k:','HandleVisibility','off')
hold off
grid on
legend('Reference tracking','Input-disturbance response','Interpreter','latex','FontSize',12)
title('Problem 1: Integral Servo (tracks and rejects)','Interpreter','latex','FontSize',15)
ylabel('$y$', 'Interpreter','latex','FontSize',16); set(get(gca,'YLabel'),'Rotation',0)
xlabel('$t$ (s)', 'Interpreter','latex','FontSize',16)
```

Problem 1: step tracking error = 0.00e+00, disturbance DC gain = 0.00e+00

### Problem 1: Integral Servo (tracks and rejects)

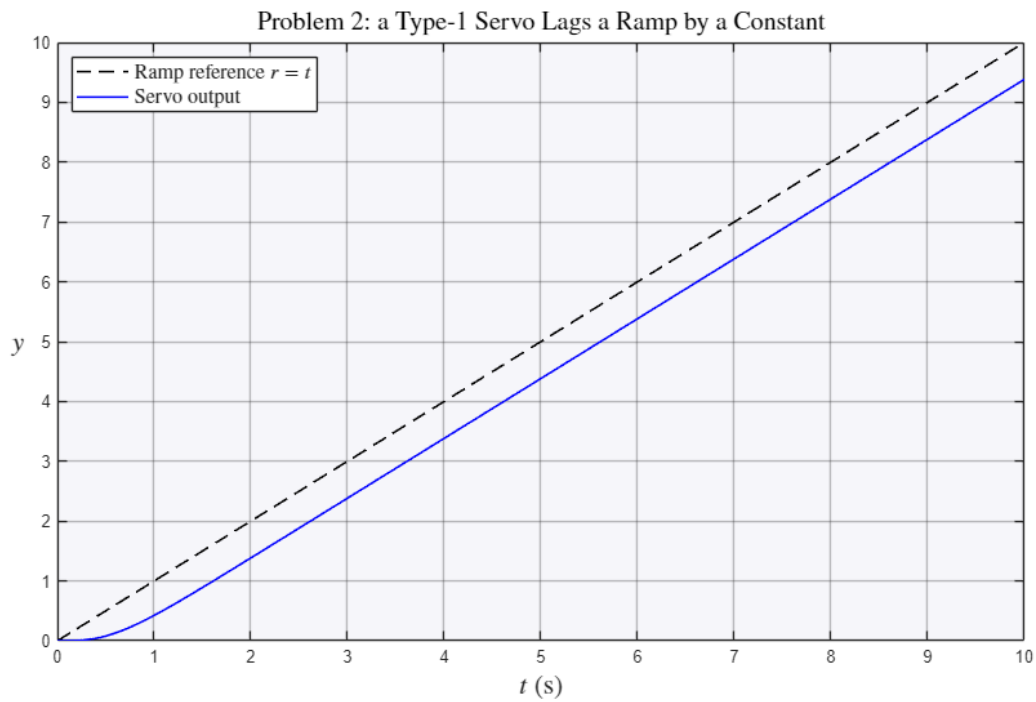


### Problem 2: what a type-1 servo can and cannot track

The integral servo above is **type 1**: zero error to a step, but a constant error to a **ramp**. Confirm it by driving the same servo with a ramp reference and measuring the steady lag.

```
tr = 0:0.01:10;
y_ramp = lsim(sys_r, tr, tr);          % ramp reference r(t) = t
figure
plot(tr, tr, 'k--', tr, y_ramp, 'b', 'LineWidth', 1.2)
grid on
legend('Ramp reference  $r=t$ ', 'Servo output', 'Interpreter', 'latex', 'FontSize', 12, 'Location', 'northwest')
title('Problem 2: a Type-1 Servo Lags a Ramp by a Constant', 'Interpreter', 'latex', 'FontSize', 15)
ylabel('$y$', 'Interpreter', 'latex', 'FontSize', 16); set(gca, 'YLabel', 'Rotation', 0)
xlabel('$t$ (s)', 'Interpreter', 'latex', 'FontSize', 16)
fprintf('Problem 2: steady ramp-tracking lag = %.4f\n', tr(end) - y_ramp(end))
```

Problem 2: steady ramp-tracking lag = 0.6167



### Try it yourself

- Add a second integrator (a type-2 servo) and notice the ramp error vanish too – at the cost of a harder-to-stabilize loop.
- Move the augmented poles faster and watch both tracking and disturbance rejection speed up together.